Simultaneous Estimation of Mutual Coupling Matrix and DOAs for UCA and ULA

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***Abstract*— A structured Least Square (LS) method for simultaneous estimation of the mutual coupling matrix (MCM) and directions of arrival (DOAs) of signal sources for uniform linear array (ULA) and uniform circular array (UCA) is presented. The DOAs and MCM can be simultaneously estimated by the proposed method when observations of at least two different DOAs are available. This method significantly simplifies the procedure of mutual coupling compensation and it can also be applied for the calibration of ULA and UCA. Simulation results demonstrate the efficiency and accuracy of the proposed method.**

***Index Terms*— Antenna array mutual coupling, DOA estimation**

I. Introduction

Antenna arrays have wide applications in wireless communication, radar, signal intelligence, etc. The performance of an array signal processing method may degrade significantly when the effect of the mutual coupling is ignored [1]. Mutual coupling distorts the linear phase front of the incoming signal, significantly degrading performance.

Mutual coupling effects are especially crucial in DOA estimation. To relieve the effects of the mutual coupling in DOA estimation, various compensation methods were proposed [2]-[5]. However, most of them were not so practical in real applications. In [2], the degradation of the performance was observed in the direct data domain algorithm. The method of moment (MOM) was used to calculate the mutual impedance matrix in antenna array. In [3], the effects of the mutual coupling were analyzed by using the open circuit voltages method in adaptive antennas. In [4], [5], the authors applied a more accurate method to calculate the mutual impedance matrix based on an estimated current distribution carries a direction reference of the incoming signal. It was shown that this method can significantly reduce the mutual coupling effect and thus lead to an improved performance of the MUSIC [6] and EPSRIT [7] DOA estimation . However, in the above mentioned methods, the computation of mutual coupling matrix (MCM) were very intensive and time-consuming.

In this paper, we propose a new method to simplify MCM computation and DOA estimation in ULA and UCA. The main advantage of the proposed method is that the MCM and DOAs can be simultaneously estimated with at least two different DOAs required. This method is can be very useful in practical applications.

The structure of an MCM can be transformed into the product of two matrices. One is a diagonal matrix in which the diagonal elements are the gains of the individual array elements. The other is a matrix whose elements are the mutual coupling coefficient between array elements. In the case of absence of gain and phase mismatching between the array elements and assuming identical impedances at the output of each elements, it has already known that for some arrays with special geometries, e.g., the well studied ULA and UCA, their MCM have special structures, symmetric Toeplitz matrix for ULA and circular matrix for UCA [3]. This property of MCM can be well exploited in the real calculation process, which greatly simplify the estimation procedure. The gain matrix of each antenna is easy to be measured. Therefore, in this paper, we assume that the gain matrix of an array is known.

II. Proposed Method for ULA

Consider a ULA with M elements. Assume that L narrow-band signal sources, centered around a known wavelength λ_0 impinging on the array from L distinct directions $\theta_1, \cdots, \theta_L$ asynchronously. For simplicity, assume that the sources and the array antennas are co-planar and the sources are in the far-field of the array.

It can be proved that for ULA, the MCM denoted by $\mathbf{A}$ is a symmetric Toeplitz matrix [8], which can be represented as

$$\mathbf{A} = \begin{bmatrix} a_1 & a_2 & \cdots & a_M \\ a_2 & a_1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_2 \\ a_N & \cdots & a_2 & a_1 \end{bmatrix} = \sum_{i=1}^{M} a_i \mathbf{E}_i \tag{1}$$

where $\mathbf{E}_i$ is a symmetric Toeplitz matrix. Its first row vector is $[0 \ \cdots \ 1 \ 0 \ \cdots \ 0]$, where the i-th element is one,others are all zero.

Using complex envelope representation, the $M \times 1$ vector received by the array can be expressed by

$$\mathbf{y}(n) = x(n)\mathbf{A}\mathbf{s}(\theta) + \mathbf{w}(n) \tag{2}$$

where $\mathbf{w}(n)$ denotes the noise vector at n-th element, $x(n)$ the incident signal on the n-th element from the direction θ, $\mathrm{s}(\theta)$ the array steering vector (ASV) of the array toward direction θ.

$$\mathbf{s}(\theta) = \left[1 \ e^{-j\frac{\pi}{2}sin(\theta)} \ \cdots \ e^{-j\frac{(M-1)\pi}{2}sin(\theta)}\right]^T \tag{3}$$

In the controlled environment of calibration phase, the observed ASV $\bar{\mathrm{s}}(\theta)$ is easy to be measured using the antenna output voltages. For L different calibration sources, we have L estimated ASVs,

$$\bar{\mathbf{s}}(\theta_i) = \mathbf{A}\mathbf{s}(\theta_i) + \mathbf{e}_i, \quad i = 1, \cdots, L \tag{4}$$

where $\mathbf{e}_i$ is the measurement error.

If the exact θ_i in the experiment are already known, with sufficient number of L experiments , the MCM $\mathbf{A}$ can be estimated by using LS method to solve the following optimization problem,

$$\hat{\mathbf{A}} = arg \min_{\mathbf{A}} \sum_{i=1}^{L} ||\bar{\mathbf{s}}(\theta_i) - \mathbf{A}\mathbf{s}(\theta_i)||_2^2 \tag{5}$$

where, $|| * ||_2$ is Euclidean norm.

Since there are M unknown parameters in MCM, and with one more experiment performed, one more unknown parameter θ_i will be included in optimization problem. Therefore, if L experiments are performed, there are $M + L$ unknown parameters in (5). For one experiment performed, there are M linear equations, same as (3), which can be obtained. In case of L experiments carried out, there are $M \times L$ linear equations available. Therefore, if $M \times L > (L + M)$, it is possible to determine the unknown parameters uniquely. That means, if $L \geq (M/(M-1))$ experiments are carried out, we can obtain L independent steering vector $\mathbf{s}(\theta_i)$, then the estimate of MCM $\mathbf{A}$ can be uniquely determined. The smallest number of L is 2.

However, for some practical experiments, the direction of signal source θ_i can not be accurately known, or totally unknown. In such cases, the estimates of MCM have errors as well as the estimated DOAs. In this paper, we propose an efficient method to estimate DOAs and MCM simultaneously.

Our problem can be stated as follows. Given a uniformly spaced linear array in the presence of mutual coupling among array elements. The MCM $\mathbf{A}$ and DOAs $\{\theta_i\}$ are all unknown. The following optimization method is used to find the estimates of MCM and DOAs.

$$\begin{aligned} J(\mathbf{A}, \theta_i) &= \sum_{i=1}^{L} ||\bar{\mathbf{s}}_i - \mathbf{A}\mathbf{s}_i||_2^2 \\ \{\hat{\theta}_i, \hat{\mathbf{A}}\} &= arg \min_{\mathbf{A}, \theta_i} J(\mathbf{A}, \theta_i) \end{aligned} \tag{6}$$

It can be proved that for fixed DOAs, the optimization problem (6) to $\mathbf{A}$ is a quadratic problem. The closed form of the solution is easy to be derived. However, the optimization problem to θ_i is nonlinear and not quadratic, so it is difficult to get its closed form solution. To simplify the optimization problem, we proposed to solve (6) using two steps. First assume that θ_i is known, then obtain the solution for $\mathbf{A}$. By substituting estimated $\mathbf{A}$ into (6), this optimization problem is simplified and only with unknown parameters θ_i. If more accurate DOA estimation is required, iterative approach can be used to estimate the MCM until solutions converge.

Substituting (1) into (6), we have

$$\begin{aligned} J(\mathbf{A}, \theta_i) &= \sum_{i=1}^{L} ||\bar{\mathbf{s}}_i - \mathbf{A}\mathbf{s}_i||_2^2 \\ &= \sum_{i=1}^{L} \bar{\mathbf{s}}_i^H \bar{\mathbf{s}}_i - \bar{\mathbf{s}}_i^H \mathbf{A}\mathbf{s}_i - \mathbf{s}_i^H \mathbf{A}^H \bar{\mathbf{s}}_i + \mathbf{s}_i^H \mathbf{A}^H \mathbf{A}\mathbf{s}_i \\ &= \sum_{i=1}^{L} \bar{\mathbf{s}}_i^H \bar{\mathbf{s}}_i - \sum_{i=1}^{L}\sum_{k=1}^{N} a_k \bar{\mathbf{s}}_i^H \mathbf{E}_k \mathbf{s}_i - \sum_{i=1}^{L}\sum_{k=1}^{N} a_k^* \mathbf{s}_i^H \mathbf{E}_k \bar{\mathbf{s}}_i \\ &\quad + \sum_{i=1}^{L}\sum_{l=1}^{N}\sum_{k=1}^{N} a_k^* a_l \mathbf{s}_i^H \mathbf{E}_k \mathbf{E}_l \mathbf{s}_i \end{aligned} \tag{7}$$

The optimal estimate of $\mathbf{A}$ is obtained by setting the derivative of J with respect to a_k^* to zero

$$\frac{\partial J(\mathbf{A}, \theta_i)}{\partial a_k^*} = -\sum_{i=1}^{L} \mathbf{s}_i^H \mathbf{E}_k \bar{\mathbf{s}}_i + \sum_{i=1}^{L}\sum_{l=1}^{N} a_l \mathbf{s}_i^H \mathbf{E}_k \mathbf{E}_l \mathbf{s}_i = 0 \tag{8}$$

We have the linear equations for each k,

$$\sum_{l=1}^{N} a_l \left[\sum_{j=1}^{L} \mathbf{s}_j^H \mathbf{E}_k \mathbf{E}_l \mathbf{s}_j \right] = \sum_{j=1}^{L} \mathbf{s}_j^H \mathbf{E}_k \bar{\mathbf{s}}_j, \quad k = 1, \cdots, N \tag{9}$$

With estimated MCM $\hat{\mathbf{A}}$, the DOAs of sources can be estimated using following optimization problem,

$$\hat{\theta}_i = arg \min_{\theta_i} \sum_{i=1}^{L} ||\bar{\mathbf{s}}_i - \hat{\mathbf{A}}\mathbf{s}(\theta_i)||^2 \tag{10}$$

III. Two Proposed Methods for UCA

We can solve the same problem in circular array by using two approaches.

A. Method I

Consider a uniform array with N identical elements and radius r. Assume that L narrow-band sources, centered around a known wavelength λ_0 impinge on the array from L distinct directions $\theta_1, \cdots, \theta_L$ asynchronously with the reference point is set at the center of the array and θ is measured with respect to the line connecting the reference point with the first array element. The n-th component of ASV is given by

$$s_n(\theta) = G_n(\theta) \exp[j \frac{2\pi r}{\lambda_0} \cos(\theta - \frac{2\pi(n-1)}{N})] \tag{11}$$

From [9], we obtain that the effect of mutual coupling between the array elements is to transform the ideal steering vector $\mathbf{S}(\theta)$ to a different steering vector $\mathbf{A}\mathbf{S}(\theta)$, where $\mathbf{A}$ is the mutual coupling matrix.

In matrix notation, the $N \times 1$ vector received by the array can be expressed by

$$\mathbf{Y} = \mathbf{X}\mathbf{A}\mathbf{S}(\theta) + \mathbf{W} \tag{12}$$

where $\mathbf{W}$ denotes the $N \times 1$ noise vector of the array, $\mathbf{X}$ the signal of k-th source as received at the array center, $\mathbf{S}(\theta)$ the ASV of the array toward direction of k-th source.

Since the antenna array is rotational symmetric, it follows that the matrix $\mathbf{A}$ is a circulant matrix given by,

$$A = \begin{bmatrix} a_0 & a_{-1} & \cdots & a_{2-N} & a_{1-N} \\ a_1 & a_0 & a_{-1} & \cdots & a_{2-N} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ a_{N-2} & \ddots & \ddots & \ddots & \vdots \\ a_{N-1} & a_{N-2} & \cdots & a_1 & a_0 \end{bmatrix} \tag{13}$$

where $a_{-k} = a_{N-k}$ for $1 \leqslant k \leqslant N-1$.

The mutual coupling matrix $\mathbf{A}$, which can also be represented as follows:

$$\mathbf{A} = \sum_{i=0}^{N-1} a_i \mathbf{H_i} \tag{14}$$

where $\mathbf{H}_i$ is the i-th power of cyclic permutation matrix. $\mathbf{H}$ given by

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \tag{15}$$

Same as in linear array, we can easily obtain the observed ASV $\bar{\mathbf{s}}(\theta)$ using the antenna output voltages. For L different calibration sources, we have L observed ASVs. From L each of DOAs, θ_i $(i = 1, \cdots, L)$, we can obtain the estimated ASVs,

$$\bar{\mathbf{s}}(\theta_i) = \mathbf{A}\mathbf{s}(\theta_i) + \mathbf{e}_i, \quad i = 1, \cdots, L \tag{16}$$

where $\mathbf{e}_i$ is the measurement error.

We assume that exact θ_i are already known, with sufficient number of L experiments, the mutual coupling matrix can be estimated as follows:

$$\hat{\mathbf{A}} = arg \min_{\mathbf{A}} \sum_{i=1}^{L} ||\bar{\mathbf{s}}(\theta_i) - \mathbf{A}\mathbf{s}(\theta_i)||_2^2 \tag{17}$$

B. Method II

In [10], Davies proposes a method to transform the outputs at the antenna elements of UCA to a virtual array. The steering vector of the virtual array is Vandermonde matrix, or approximately so. Following the approach outlined above, we shall transform the N dimensional circular array to a M dimensional virtual array, $M < N$. In [11], the transformation matrix $\mathbf{T}$ is defined by

$$\mathbf{T} = \mathbf{JF} \tag{18}$$

where $\mathbf{J}$ is $M \times M$ diagonal matrix, for $m = 1, \cdots, M$,

$$\mathbf{J} = diag \left\{ \frac{1}{j^{m-1-\frac{(M-1)}{2}} \sqrt{N} J_{m-1-\frac{(M-1)}{2}} \left(\frac{2\pi r}{\lambda}\right)} \right\} \tag{19}$$

and $\mathbf{F}$ is $M \times N$ matrix,

$$\mathbf{F} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & e^{-j2\pi 1\cdot(M-1)/2N} & \cdots & e^{-j2\pi(N-1)\cdot(M-1)/2N} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & e^{-j2\pi 1\cdot 2/2N} & \cdots & e^{-j2\pi(N-1)\cdot 2/2N} \\ 1 & 1 & \cdots & 1 \\ 1 & e^{j2\pi 1\cdot 2/2N} & \cdots & e^{j2\pi(N-1)\cdot 2/2N} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & e^{j2\pi 1\cdot(M-1)/2N} & \cdots & e^{j2\pi(N-1)\cdot(M-1)/2N} \end{bmatrix} \tag{20}$$

Let $\widetilde{\mathbf{S}}(\theta)$ denote the ASV of the virtual array,

$$\widetilde{\mathbf{S}}(\theta) \cong \mathbf{TS}(\theta) \cong \mathbf{JFS}(\theta) \tag{21}$$

so, $\widetilde{\mathbf{S}}(\theta)$ is the M dimensional vector,

$$\widetilde{\mathbf{S}}(\theta) = [e^{-j\theta(M-1)/2}, \cdots, e^{-j\theta}, 1, e^{j\theta}, \cdots, e^{j\theta(M-1)/2}]^T \tag{22}$$

Thus we can use $\widetilde{\mathbf{S}}(\theta)$ to take the place of $\mathbf{S}(\theta)$ in (4). Then we can solve the minimization problem in (5) as in II.

IV. Numerical Study

In this section, numerical examples are presented to demonstrate the proposed approach. We consider a uniformly spaced linear array consisting of 8 monopoles in z-direction placed along x axis. The dimensions of the monopole elements are: length = 3.0 cm and wire radius = 0.3 mm. They are placed over a large ground plane and connected to a 50 Ω load and inter-element spacing is d = 6.25 cm (half wavelength at 2.4 GHz). The array is aligned along the x axis with the monopole elements parallel to the z axis. The incoming signal is sinusoidal plane wave with vertical polarization, amplitude = 1 V , frequency = 2.4 GHz, and additive thermal noise about -20 dB. In the experiment, the mutual coupling matrix is constructed using the a simulated Toeplitz matrix which is obtained from simulations in our previous work [5]. Assuming that we have three observations available, the true DOAs are $5°$, $25°$ and $45°$, respectively. In Fig. 1, the DOA estimates at each iteration is shown. It is clear that the converged DOA estimates have very small estimation error. Moreover, the estimated MCM elements are shown in Fig. 2. The estimation errors of MCM elements are also very small.

In the second experiment,We consider a uniformly spaced circular array consisting of 9 monopoles. The UCA of radius = 0.5 λ, was employed for the simulations, where λ is the signal wavelength. The dimensions of the monopole elements are: length d = 3.31 cm and wire radius = 0.3 mm and each connected to a 50 Ω load. The signal environment consists of two signals. The incoming signal is sinusoidal plane wave with vertical polarization, amplitude = 1 V, frequency = 2.4 GHz, and additive thermal noise about -20 dB. In the experiment, the mutual coupling matrix is constructed using the a simulated circulant matrix which is obtained from simulations in our previous work [5], and then transformed by circular method II shown in this paper. Assuming that we have three observations available, the true DOAs are $30°$, $45°$ and $60°$, respectively.

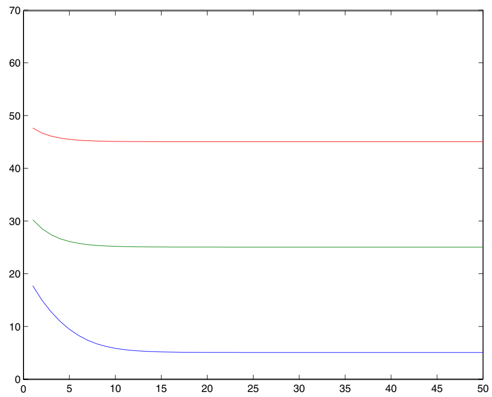

Fig. 1. DOA estimates versus iteration.

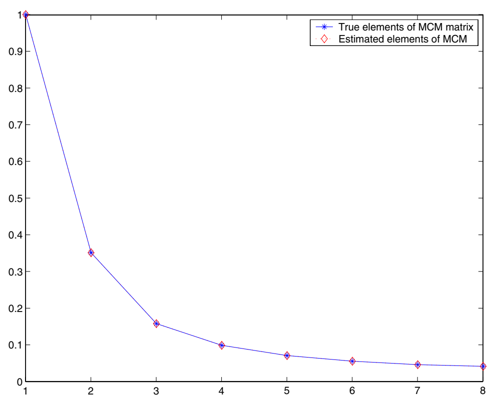

Fig. 2. MCM elements estimated after 20 iterations.

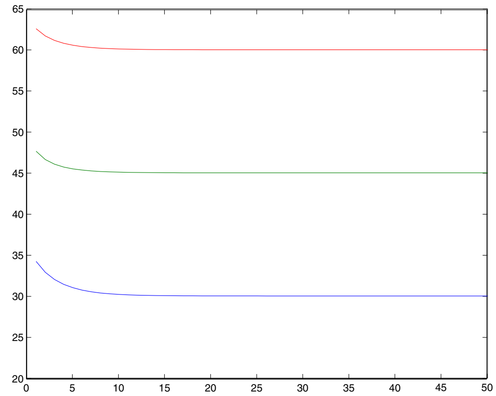

Fig. 3. DOA estimates versus iteration.

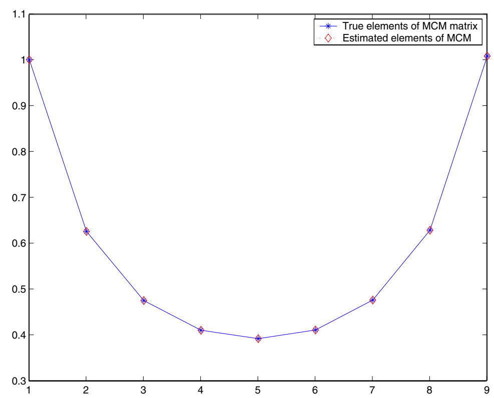

Fig. 4. MCM elements estimated after 20 iterations.

In Fig. 3, the DOA estimates at each iteration is shown. It is clear that the converged DOA estimates have very small estimation error. Moreover, the estimated MCM elements are shown in Fig. 4. The estimation errors of MCM elements and DOAs are also very small, this shows that the methods used in this paper are very effective and easy to implemented in real arrays.

V. Conclusion

A structured LS method is proposed to simultaneously estimate the MCM of a ULA or a UCA and the DOAs in the presence of mutual coupling and thermal noise. It was found this method can estimate the DOA and MCM simultaneously when at least two DOAs are available. The proposed method is also useful in calibration of ULA and UCA when the DOAs of calibration sources were not known. Two examples validate the proposed method.

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